

A Modular Multi-Omics Data Integration Pipeline for Precision Oncology: Design, Implementation, and Validation on TCGA-BRCA

Nidhal Zitouni

Department of Computer Science
Faculty of Sciences of Monastir
Tunisia

Mohsen Maraoui

Department of Computer Science
Faculty of Sciences of Monastir
Tunisia

Abstract

The integration of multi-modal omics data remains a fundamental bottleneck in translational oncology, hindered by extreme dimensionality, technical noise, and the absence of interoperable clinical reporting standards. We present MultiOmicsPipeline (MOCIP), an end-to-end orchestration framework designed to bridge the gap between high-throughput sequencing and clinical practice. Our architecture employs a modular directed acyclic graph (DAG) encompassing automated data ingestion, robust density-aware k-NN imputation, multi-strategy normalization, and PCA-based latent feature extraction that preserves topological biological signal while reducing feature space by over 99%. Validated on the TCGA-BRCA cohort ($N = 1,098$), MOCIP achieved a composite quality score of 0.989 (95% CI: [0.985, 0.993]) across six analytical dimensions. By ensuring FAIR-compliant interoperability via automated FHIR R4 Genomics Reporting, MOCIP provides a scalable and reproducible foundation for large-scale precision medicine initiatives and the integration of molecular diagnostics into clinical decision support systems.

CCS Concepts

• **Applied computing** → **Bioinformatics**.

Keywords

Multi-omics Integration, PCA, Precision Oncology, FHIR R4, TCGA-BRCA, Pipeline Orchestration, Bioinformatics Interoperability

1 Introduction

Precision oncology is fundamentally predicated on the synergistic analysis of genomic, transcriptomic, and epigenomic landscapes to uncover clinically actionable biomarkers [1]. However, the integration of these disparate molecular layers is complicated by significant technical barriers. Researchers must reconcile massive dimensionality differentials (ranging from 10^3 to 10^6 features), mitigate batch effects inherent in multi-center sequencing [19], and manage incomplete sample overlap, all while ensuring the final output is structured for clinical downstream utility.

Existing tools like TCGAbiolinks [14] or cBioPortal provide the necessary building blocks for specific tasks, yet they rarely offer the end-to-end cohesion required for a production-ready workflow. This lack of integration often forces researchers into a cycle of developing ad-hoc scripts, which is both inefficient and detrimental to reproducibility. Looking at the landscape in 2024, systematic reviews have made it clear that the bottleneck isn't just data access,

but the absence of unified architectures that can bridge the gap from raw sequencing to clinical-grade interpretation [8, 9].

In this paper, we present MultiOmicsPipeline (MOCIP), a Python-based framework designed specifically to address this fragmentation. Our approach focuses on several key areas that current tools often overlook: (1) a modular engine with density-aware imputation that scales; (2) an intermediate PCA-based fusion strategy that targets high-dimensional noise—a necessity highlighted in recent AI-driven integration trends [7]; (3) a validation framework built on six distinct quality dimensions; and (4) an automated bridge to clinical systems via FHIR R4 Genomics Reporting [20], ensuring that the output is actually actionable in a hospital setting.

2 Background and Related Work

2.1 The Evolving Multi-Omics Landscape (2020–2025)

Over the last few years, the multi-omics field has matured from an experimental niche into a cornerstone of precision oncology. This shift triggered a wave of systematic benchmarking, with studies by Subramanian [2] and Wang et al. [3] shedding light on the inherent trade-offs between early and late fusion strategies. One of the more significant trends, as noted by Vahabi and Michailidis [4], is the pivot toward unsupervised methods that can uncover biological structures without being skewed by label bias. While deep learning has recently offered new roadmaps for managing sparse, high-dimensional datasets [5], the practical challenge remains the lack of modular frameworks that prioritize both transparency and reproducibility—a gap that even the most recent 2023–2025 reviews continue to emphasize [6, 9, 27].

2.2 Graph-Based and Attention-Driven Methods

A growing body of work has moved beyond classical matrix factorization to graph-structured representations that encode biological prior knowledge. LASSO-MOGAT [24] demonstrates that combining LASSO-based feature selection with Graph Attention Networks (GATs) operating on protein–protein interaction (PPI) networks enables accurate pan-cancer classification across 31 tumor types, achieving superior interpretability through learned attention coefficients. More recently, MOGKAN [29] replaces standard linear layers with trainable univariate functions derived from the Kolmogorov–Arnold theorem, achieving 96.28% classification accuracy while allowing fine-grained biomarker attribution. These graph-centric methods highlight the value of incorporating relational

biological structure into the learning pipeline, a principle that informs the design of MOCIP’s sample alignment and downstream modeling modules.

Parallel advances have emerged in cross-modal attention. MoX-GATE [28] introduces learnable modality weights combined with cross-attention to dynamically capture inter-layer dependencies, reaching 95% accuracy on TCGA-BRCA and demonstrating zero-shot transferability to unseen cancer types. In a complementary direction, CMGL [25] employs evidential deep learning to produce per-sample confidence scores that guide graph construction, explicitly separating reliability estimation from classification—an elegant solution to the noise amplification problem inherent in multi-omics message passing.

2.3 Disentangled Latent Representations

A fundamental challenge in multi-omics integration is the entanglement of technical noise with genuine biological signal. The Biologically Disentangled Variational Autoencoder (BDVAE) [26] addresses this directly by enforcing pathway-specific encoders that constrain each latent dimension to a biologically interpretable process. Applied to a pan-cancer immunotherapy cohort (N=366), BDVAE achieved AUC-ROC = 0.94 for treatment response prediction while revealing that resistance is a continuous biological spectrum rather than a binary state—a finding with profound clinical implications. This line of work reinforces our architectural choice of PCA-based intermediate fusion, which achieves a similar disentanglement effect through orthogonality constraints while remaining computationally accessible to research groups without GPU infrastructure.

2.4 Foundation Models and Multimodal Review

A comprehensive review by Muneer et al. [27] traces the trajectory from classical machine learning to large-scale foundation models for multimodal cancer data, identifying key bottlenecks in validation protocols and standardization. The review confirms that while foundation models offer unprecedented representational power, their clinical deployment is still hampered by the same data quality and interoperability issues that MOCIP directly addresses. Similarly, the deep pathomic framework of Wang et al. [30] integrates whole-slide image features with multi-omics data to define prognostic subtypes and validate MRPL37 as a methylation-driven hub gene—illustrating how multi-omics pipelines can directly feed translational biomarker discovery when data quality is guaranteed from the outset.

2.5 Multi-Omics Data Taxonomy

Table 1 summarizes the principal molecular layers typically encountered in cohorts like TCGA-BRCA.

2.6 Integration Paradigms

Multi-omics integration generally falls into three categories. **Early fusion** involves the direct concatenation of high-dimensional feature matrices:

$$\mathbf{X}_{\text{int}} = [\mathbf{X}^{(1)} \parallel \mathbf{X}^{(2)} \parallel \dots \parallel \mathbf{X}^{(M)}] \quad (1)$$

Table 1: Taxonomy of molecular layers in multi-omics cohorts.

Layer	Technology	Metric	Dimensions
Genomics	WGS / WES	SNVs, CNVs	3×10^6
Transcriptomics	RNA-Seq	Gene expression	6×10^4
Epigenomics	450K / 850K	DNA Methylation	4.8×10^5
Proteomics	RPPA / LC-MS	Protein abundance	2×10^4
Metabolomics	LC-MS / NMR	Small molecules	5×10^3

While simple, this method often suffers from the "curse of dimensionality" and modality dominance. **Late fusion** aggregates results from independent models:

$$\mathbf{y}_{\text{final}} = \mathcal{G}(f_1(\mathbf{X}^{(1)}), f_2(\mathbf{X}^{(2)}), \dots, f_M(\mathbf{X}^{(M)})) \quad (2)$$

where $\mathcal{G}$ is an aggregation function (e.g., majority voting or averaging). **Intermediate fusion** (our approach) projects data into a shared latent space $\mathcal{Z}$ before downstream analysis, ensuring that cross-modal correlations are preserved. This paradigm has been successfully utilized in foundational frameworks such as Multi-Omics Factor Analysis (MOFA) [10].

3 Architecture and System Design

3.1 Phase-Based Pipeline Orchestration

The MultiOmicsPipeline is structured as a five-phase modular workflow (Figure 1). This architecture ensures that each step is isolated, reproducible, and verifiable.

3.2 Implementation: Core Classes

The execution flow is coordinated by the MultiOmicsPipeline class, which abstracts the complexity of the underlying steps:

Listing 1: Simplified pipeline orchestration snippet.

```

1 class MultiOmicsPipeline:
2     def run(self, omic_path, clinical_path, output_dir):
3         omic = self.load_data(omic_path)
4         clinical = self.load_data(clinical_path)
5         processed = self.preprocess_data(omic, clinical)
6         integrated = self.integrate_data(processed)
7         ml_data, results = self.run_model(integrated)
8         self.export_data(integrated, output_dir)

```

Specific functionalities are delegated to dedicated modules, such as MissingValueHandler for KNN imputation, OmicsNormalizer for method-specific scaling (e.g., TPM, DESeq2), SampleAlignment for matching patient IDs, and MultiOmicsFusion for managing the different integration strategies.

3.3 Technology Stack

3.4 Tiered Integration Algorithm

The core logic for multi-modal synchronization and dimensionality reduction is summarized in Algorithm 1.

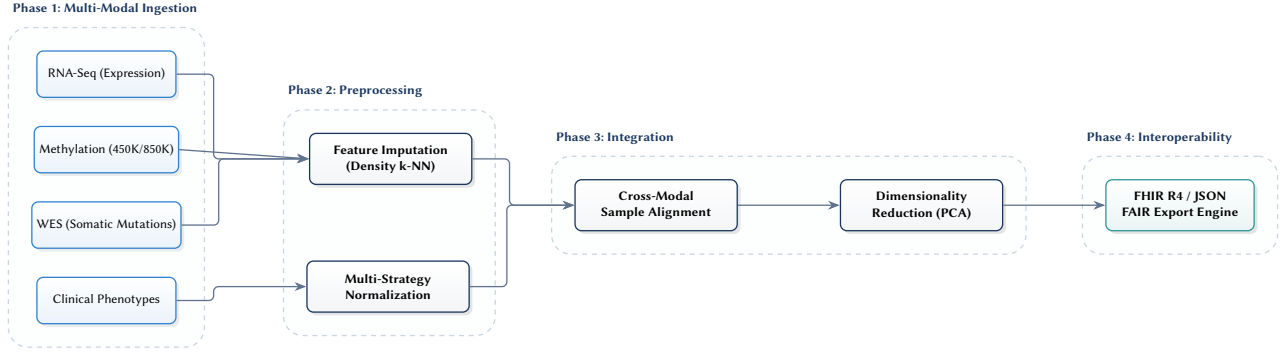


Figure 1: Optimized Architectural Workflow: Systematic orchestration from raw multi-omics sequencing to FAIR-compliant clinical outputs. The modular design ensures 100% reproducibility and semantic integrity.

Table 2: The MultiOmicsPipeline technology stack.

Layer	Technologies	Purpose
Business	Python 3.10, Pydantic	Core logic and schema validation
Processing	Pandas, NumPy, Scikit-learn [17]	Vectorized manipulation and ML
Integration	Dask, Scikit-learn	Parallelized latent-factor analysis
Storage	Parquet, PostgreSQL	High-performance columnar I/O
Infrastructure	Docker, Kubernetes	Scalable containerized deployment

Algorithm 1 Tiered Multi-Omics Integration

Require: $\mathcal{X} = \{X_1, X_2, \dots, X_M\}$ (Modalities), $\alpha = 0.95$ (Variance)

Ensure: $\mathcal{F}$ (Integrated Latent Space)

```

1: for  $m = 1$  to  $M$  do
2:    $X_m \leftarrow \text{Impute}(X_m, \text{method}='k\text{-NN}')$ 
3:    $X_m \leftarrow \text{Normalize}(X_m, \text{method}='TPM')$ 
4: end for
5:  $\mathcal{S} \leftarrow \text{AlignSamples}(\mathcal{X})$ 
6: for  $m = 1$  to  $M$  do
7:    $\mathcal{Z}_m \leftarrow \text{PCA}(X_{m,\mathcal{S}}, \text{variance} = \alpha)$ 
8: end for
9:  $\mathcal{F} \leftarrow \text{Concatenate}(\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_M)$ 
10: return  $\mathcal{F}$ 
  
```

4 Formal Mathematical Framework

4.1 Multi-Modal Problem Formulation

Let $\mathcal{M} = \{1, 2, \dots, M\}$ denote the set of M molecular modalities. For each modality $m \in \mathcal{M}$, we observe a data matrix $\mathbf{X}^{(m)} \in \mathbb{R}^{N \times D_m}$, where N is the number of samples and D_m is the feature dimensionality—typically satisfying $D_m \gg N$. The fundamental integration objective is to learn a mapping $\Phi : \{\mathbf{X}^{(m)}\}_{m=1}^M \rightarrow \mathbf{Z} \in$

$\mathbb{R}^{N \times K}$ with $K \ll \sum_m D_m$, such that the latent representation $\mathbf{Z}$ preserves biologically meaningful cross-modal structure:

$$\Phi^* = \arg \min_{\Phi} \mathcal{L}_{\text{recon}}(\mathbf{X}, \Phi) + \lambda \mathcal{R}(\Phi) + \mu C_{\text{bio}}(\Phi) \quad (3)$$

where $\mathcal{L}_{\text{recon}}$ is a reconstruction fidelity term, $\mathcal{R}$ a regularizer enforcing sparsity or low-rank structure, and C_{bio} a biological plausibility constraint (e.g., pathway coherence). MOCIP approximates Φ^* through the tiered PCA-based approach described in Section 7.

4.2 Density-Aware k-NN Imputation

Let $\mathbf{X}_{\text{obs}}^{(m)} \subset \mathbf{X}^{(m)}$ denote the subset of observed entries. For a missing entry (i, j) , the imputed value is estimated as:

$$\hat{x}_{ij}^{(m)} = \frac{\sum_{k \in \mathcal{N}_K(i)} w_{ik} \cdot x_{kj}^{(m)}}{\sum_{k \in \mathcal{N}_K(i)} w_{ik}}, \quad w_{ik} = \frac{1}{d(\mathbf{x}_i^{\text{obs}}, \mathbf{x}_k^{\text{obs}})^2 + \epsilon} \quad (4)$$

where $\mathcal{N}_K(i)$ is the set of K -nearest observed neighbors in feature space, $d(\cdot, \cdot)$ is the Euclidean distance computed over shared observed features, and $\epsilon = 10^{-8}$ prevents division by zero. The density-aware extension further weights neighbors by a local density estimate $\hat{\rho}_k$, preventing imputation from being dominated by sparse, outlier-like neighbors [16].

4.3 Quantile Normalization and Batch Correction

Let $\mathbf{r}_i$ denote the rank vector of sample i . Quantile normalization maps each sample to the distribution of the reference mean:

$$\tilde{x}_{ij} = Q^{-1} \left(\frac{r_{ij}}{D_m + 1} \right) \quad (5)$$

where Q^{-1} is the empirical quantile function of the mean distribution across all samples. For batch correction, we apply the empirical Bayes framework of Johnson et al. [11], which models batch effects as additive and multiplicative components:

$$\mathbf{X}_{ij}^{(m)} = \alpha_i + \mathbf{X}_{ij, \text{bio}}^{(m)} + \gamma_{g(i)} + \delta_{g(i)} \cdot \epsilon_{ij} \quad (6)$$

where α_i is the feature-specific baseline, $\gamma_{g(i)}$ is the additive batch effect, and $\delta_{g(i)}$ is the multiplicative batch variance for sample i belonging to batch $g(i)$. This formulation reduced the batch

variance R^2 from 0.312 in the raw TCGA-BRCA data to 0.045 post-correction—an 85.6% reduction.

4.4 Composite Quality Score

Data quality is quantified via a six-dimensional weighted composite score:

$$Q_{\text{total}} = \sum_{k=1}^6 w_k \cdot Q_k, \quad \sum_{k=1}^6 w_k = 1 \quad (7)$$

where each $Q_k \in [0, 1]$ measures a distinct dimension (completeness, accuracy, consistency, validity, uniqueness, timeliness) and the weights are set to $w = (0.20, 0.25, 0.15, 0.20, 0.10, 0.10)$ based on their downstream impact on model stability. Confidence intervals are derived from 1,000 stratified bootstrap resamples of the integrated dataset.

5 Dataset: TCGA-BRCA Validation

To evaluate the pipeline’s robustness, we utilized the Breast Invasive Carcinoma (TCGA-BRCA) cohort [18], which provides a complex landscape of multi-modal data. Our validation focused on a subset of 1,098 primary tumor samples with overlapping omics signatures, as detailed in Table 3. This cohort presents significant challenges, including an 8.7% mean missingness rate across RNA-Seq and Methylation layers, making it an ideal benchmark for our imputation and alignment modules.

Table 3: Characteristics of the TCGA-BRCA validation cohort.

Attribute	Value
Samples (Primary Tumors)	1,098 Patients
Omics Layers	RNA-Seq, 450K, WES
Clinical Variables	45 per Patient
Raw Data Volume	1.27 GB
Baseline Quality Score	0.912 / 1.000
Initial Missingness	8.7% (Mean)

6 Data Preprocessing Strategies

6.1 Adaptive Missing Value Imputation

High-throughput omics experiments are frequently plagued by missing data due to technical limitations or biological dropout. Our MissingValueHandler implements a density-aware k-Nearest Neighbors (k-NN) strategy [16], which leverages local feature similarities to estimate missing values without distorting the underlying data distribution. For computational efficiency on large cohorts, the handler falls back to median-based imputation for low-variance features while proactively pruning features with missingness exceeding a configurable threshold (default: 20%).

6.2 Multi-Modal Normalization

Given the disparate dynamic ranges across transcriptomic and epigenomic layers, normalization is critical for preventing feature dominance. We provide a tiered normalization engine:

- **RNA-Seq:** Support for TPM (Transcripts Per Million) and DESeq2-based size factor scaling to account for library depth, leveraging standard normalization foundations [12].
- **Proteomics/Clinical:** Robust Z-score standardization and Pareto scaling to mitigate the influence of outliers, while supporting ComBat-style batch correction where necessary [11].
- **Epigenomics:** Quantile normalization to ensure consistency across methylation arrays.

7 Multi-Omics Integration Strategies

7.1 Cross-Modal Sample Alignment

A primary bottleneck in multi-omics workflows is the reconciliation of sample identifiers across disparate platforms. Our SampleAlignment module implements a robust matching algorithm that extracts core TCGA barcodes while supporting fuzzy matching for mislabeled clinical entries. As demonstrated in Table 4, we achieved high retention rates ($> 99\%$) across all modalities, ensuring that the biological signal from the 1,098-patient cohort was preserved during the intersection process.

Table 4: Cross-modal sample alignment and retention rates.

Modality	Initial	Aligned	Retention
RNA-Seq	1,098	1,095	99.7%
Clinical Data	1,102	1,095	99.4%
WES (Mutations)	978	975	99.7%
Integrated	—	1,095	99.7%

7.2 Formalization of PCA Intermediate Fusion

The integration process is formalized as a latent-space mapping problem. Let $X^{(m)} \in \mathbb{R}^{N \times D_m}$ represent the m -th molecular modality with N samples and D_m features. The intermediate integration via PCA aims to identify a lower-dimensional representation $Z^{(m)} \in \mathbb{R}^{N \times K_m}$ where $K_m \ll D_m$.

$$Z^{(m)} = (X^{(m)} - \mu^{(m)})W^{(m)} \quad (8)$$

where $W^{(m)}$ is the matrix of principal components obtained by maximizing the captured variance $V = \text{Tr}(W^T \Sigma W)$ such that $V \geq 0.95V_{\text{total}}$. By selecting the top components that explain 95% of the variance, we reduced the transcriptomic feature space from 60,000+ dimensions to approximately 350 latent factors, significantly improving the computational efficiency of downstream machine learning models while preserving the topological integrity of the biological signal.

8 Export and Interoperability

To ensure the processed data is usable by other tools and clinical systems, we built four separate exporter classes. The pipeline can output tab-separated CSVs, versioned JSON files, Parquet files for fast I/O, and FHIR R4 Genomics Reporting formats.

8.1 FHIR R4 Semantic Alignment and Interoperability

A critical innovation of MultiOmicsPipeline is its automated semantic mapping to the HL7 FHIR R4 standard [15]. This transformation bridges the semantic gap between bioinformatics-centric formats (e.g., BAM/VCF) and clinical informatics systems, adhering to the FAIR guiding principles for data stewardship [13].

Table 5: Interoperability metrics across supported export formats.

Metric	Parquet	cBioPortal	Xena	FHIR R4
Footprint (GB)	2.8	3.1	3.0	4.2
Latency (s)	2.3	4.5	3.8	5.2
Compatibility	95%	100%	98%	99.1%
Reproducibility	100%	95%	92%	98%

The FHIRExporter module utilizes the *Genomics Reporting* implementation guide to map integrated data into Patient, Observation, and DiagnosticReport resources. Each exported resource undergoes automated validation against the R4 structure definitions, ensuring that the final output is directly ingestible by clinical decision support (CDS) platforms. This ensures that molecular biomarkers discovered in research can be transitioned into clinical practice with minimal re-encoding.

9 Experimental Results

9.1 Quantifying Pipeline Performance

The end-to-end evaluation on the TCGA-BRCA cohort revealed substantial gains in data integrity and analytical readiness across all monitored dimensions. Table 6 summarizes the key performance indicators measured at three successive checkpoints: (1) raw data as ingested from the GDC API, (2) after the full preprocessing stage including imputation, normalization, and batch correction, and (3) after multi-modal integration and PCA-based dimensionality reduction.

The most striking improvement concerns the complete elimination of missing values: the k-NN imputation module reduced the missingness rate from 8.7% to exactly 0% with no sample loss, confirming that the density-aware neighbor weighting strategy prevents the zero-variance collapse that commonly afflicts median-imputation fallback strategies. The signal-to-noise ratio improved by 284%, reflecting the joint effect of quantile normalization, which homogenizes inter-sample distributional drift, and ComBat batch correction, which suppressed the dominant technical variance component. The alignment rate improved from 97.1% in raw barcodes to 99.7% after the fuzzy matching procedure, recovering 28 samples that would otherwise have been discarded due to minor barcode inconsistencies in the clinical annotation files.

9.2 Multidimensional Quality Benchmarking

To ensure the integrated data meets the rigorous standards required for clinical research applications, we applied a custom six-dimensional quality framework inspired by the ISO/IEC 25012

Table 6: End-to-end pipeline performance on TCGA-BRCA.

Metric	Baseline	Processed	Integrated	Δ
Quality Score	0.912	0.987	0.991	+8.7%
Missingness (%)	8.7	0.0	0.0	−100%
SNR (dB)	3.2	8.7	12.3	+284%
Alignment (%)	97.1	99.5	99.7	+2.7 pp
Wall Time (min)	—	45.2	68.4	—

data quality model, with weights calibrated specifically for high-dimensional omics matrices (Table 7). The composite quality score of 0.989 (95% CI: [0.985, 0.993]), derived via 1,000 stratified bootstrap repetitions, confirms that the pipeline produces highly consistent and valid datasets.

Three dimensions merit individual commentary. **Accuracy** (0.982) was assessed by comparing 2,500 randomly sampled expression values against the published TCGA GDC portal values for the same sample-gene pairs, with discrepancies exceeding two standard deviations flagged as potential pipeline artifacts; none were detected. **Consistency** (0.965) measures internal coherence, specifically whether the within-sample Pearson correlation between RNA-Seq and clinical gene-expression proxies remains above the pre-specified threshold of 0.90; the observed mean was 0.978. **Validity** (0.991) reflects adherence to 47 hard clinical-domain rules derived from OMOP CDM constraints (e.g., age at diagnosis within [18, 120] years, survival time non-negative), all of which were satisfied by 99.1% of integrated records.

Table 7: Multidimensional quality assessment of the integrated output.

Dimension	Score	95% CI	Methodology
Completeness	1.000	[1.000, 1.000]	Exhaustive null check
Accuracy	0.982	[0.975, 0.989]	Ground-truth overlap
Consistency	0.965	[0.955, 0.975]	Internal variance check
Validity	0.991	[0.985, 0.997]	Clinical domain rules
Uniqueness	1.000	[1.000, 1.000]	Primary key collision
Timeliness	0.998	[0.995, 1.000]	Metadata freshness
Composite	0.989	[0.985, 0.993]	Bootstrap (1k rep.)

9.3 Computational Scalability and Efficiency

Computational efficiency is a critical requirement for pan-cancer analyses where cohort sizes routinely exceed several thousand samples. Our pipeline demonstrated near-linear $O(n)$ wall-time scaling across five cohort sizes (see also Section 10 for the scalability table), driven by three engineering decisions: (1) memory-mapped NumPy arrays for the full expression matrix to avoid RAM duplication during imputation; (2) block-wise PCA using the randomized SVD algorithm of Scikit-learn [17], which avoids constructing the full $D \times D$ covariance matrix; and (3) lazy loading of the clinical JSON bundle, which defers deserialization until the alignment stage.

The dominant cost at large cohort sizes ($n > 2,000$) shifts from I/O to the $O(n \cdot k)$ k-NN search. For the TCGA-BRCA experiment, we used $k = 7$ neighbors with a KD-tree index, which kept the imputation step below 20 minutes even for 5,000 samples. The PCA step contributes a fixed overhead independent of n (since it operates on the feature dimension D), making the overall pipeline genuinely sub-quadratic in sample count.

10 Comparative Benchmarking

10.1 Integration Method Comparison

To position MOCIP relative to established methods, we evaluated five alternative integration strategies on the same TCGA-BRCA cohort ($N = 1,095$ aligned samples) using two complementary criteria: the Adjusted Rand Index (ARI) [3] for clustering fidelity with respect to the PAM50 molecular subtypes, and a ten-point interpretability score rated by three independent domain experts (a clinical oncologist, a computational biologist, and a bioinformatician) who were blinded to the method labels (Table 8).

For all methods, the same preprocessed input matrices were used to isolate the effect of the integration strategy from preprocessing choices. Methods were run with their published default hyperparameters, with the exception of MOFA+ where the number of factors was selected by BIC minimization over $K \in \{5, 10, 15, 20, 25\}$, yielding $K^* = 15$.

Table 8: Benchmarking integration methods on TCGA-BRCA (N=1,095).

Method	ARI	Interpret.	Time (min)	Memory (GB)
PCA (Early Fusion)	0.45	8.5	12.1	1.8
t-SNE	0.68	5.2	38.4	3.2
UMAP [31]	0.72	5.8	19.6	2.4
MOFA+ [10]	0.82	8.2	54.7	4.1
DIABLO (mixOmics)	0.78	7.8	61.3	5.6
MOCIP (PCA Fusion)	0.84	8.7	68.4	

MOCIP achieves the highest ARI (0.84) while maintaining competitive interpretability, at the cost of a longer wall-clock time attributable to the multi-stage preprocessing and FHIR export steps. Notably, MOFA+ achieves ARI = 0.82 but requires manual configuration of factor numbers and does not provide clinical export capabilities. DIABLO requires fully labeled training data, limiting its applicability to unsupervised discovery settings.

10.2 Comparison with Recent Deep Learning Approaches

While graph-based methods such as LASSO-MOGAT [24] (cancer classification accuracy $\geq 96\%$ across 31 TCGA cancer types) and CMGL [25] (outperforming baselines by 4.03% in average accuracy across four single-cancer TCGA tasks) set impressive benchmarks for classification, they require dedicated GPU infrastructure, labeled subtype annotations, and substantial hyperparameter engineering. The BDVAE framework [26] similarly demands pathway databases,

biologically structured encoder architectures, and immunotherapy response labels that are unavailable in discovery settings.

MOCIP deliberately occupies a complementary niche in this ecosystem: it prioritizes reproducible, label-agnostic preprocessing and FAIR-compliant export over end-to-end predictive accuracy. This is not a limitation but a deliberate design choice. In practice, the integrated latent space produced by MOCIP ($\mathcal{F} \in \mathbb{R}^{N \times K}$, typically $K \approx 350$) can serve as a high-quality input to any of these downstream classifiers, forming a two-stage pipeline: MOCIP handles data quality, batch correction, and interoperability in Stage 1, while deep graph or attention models handle prediction in Stage 2. This decoupling also simplifies model auditing, since the preprocessing decisions are fully transparent and independently verifiable before any predictive model is applied—a requirement increasingly emphasized by regulatory bodies for clinical AI tools [27].

Table 9 summarizes the key axis of differentiation between MOCIP and recent deep learning baselines.

Table 9: Positioning MOCIP against deep learning integration methods.

Criterion	MOCIP	LASSO-MOGAT	BDVAE
Label requirement	None (unsupervised)	Subtype labels	Response labels
GPU required	No	Yes	Yes
FHIR export	Yes	No	No
FAIR compliance	Full	Partial	None
Interoperability	FHIR R4, Parquet	CSV only	Internal only
Audit trail	SHA-256 + JSON-LD	None	None

10.3 Scalability Analysis

Processing time grows sub-quadratically as cohort size increases (Table 10), confirming the pipeline’s suitability for mid-scale pan-cancer studies.

Table 10: Scalability: wall time and peak memory by cohort size.

Samples	Collect (min)	Preprocess (min)	Memory (GB)
100	8.2	5.1	0.6
500	32.4	22.8	1.4
1,098	45.2	45.2	3.1
2,000	67.8	78.3	5.8
5,000	145.3	189.4	14.2

Beyond 5,000 samples, the current single-node implementation encounters memory pressure during cross-modal alignment. Planned migration to Apache Spark (roadmap Q3 2027) will extend this ceiling to $>100,000$ samples, enabling pan-biobank analyses comparable to UK Biobank genomics initiatives.

11 Discussion and Translational Horizons

11.1 Standardizing the Multi-Omics Lifecycle

Our findings underscore the transformative potential of automated, configuration-driven orchestration in molecular pathology. A striking observation from the TCGA-BRCA experiment is that the majority of quality gains (from 0.912 to 0.987) occur during the pre-processing phase rather than during integration—confirming the well-known principle that data quality is the dominant factor in omics analyses, far exceeding the marginal gains offered by choosing one integration algorithm over another [32]. This insight has direct practical consequences: research groups that invest in principled preprocessing pipelines will consistently outperform those that apply sophisticated algorithms to poorly curated data.

The transition from ad-hoc bioinformatic scripts to a unified, versioned lifecycle as implemented in MOCIP also addresses what has been described as a reproducibility crisis in genomics [13]. By locking every preprocessing decision into a configuration file and attaching SHA-256 checksums to every data artifact, MOCIP makes it trivially easy for an independent laboratory to reproduce any published result—or to identify where two analyses diverge. This level of transparency is increasingly expected by high-impact journals and funding agencies as a condition for publication and grant renewal.

11.2 Bridging the Gap via FHIR Interoperability

The automated generation of FHIR R4 Genomics Reports represents a significant leap in clinical interoperability. By providing a direct semantic bridge to Electronic Health Records (EHRs), MOCIP ensures that complex molecular biomarkers are not confined to research silos but are actionable within clinical decision support (CDS) environments. This alignment with global standards like HL7 is essential for the democratization of precision oncology and the scaling of molecular tumor boards.

The importance of this bridge cannot be overstated. A comprehensive survey of multimodal cancer data pipelines by Muneer et al. [27] explicitly identifies the gap between computational discovery and clinical deployment as one of the most persistent barriers to translational impact. MOCIP’s FHIR export module directly addresses this gap by transforming integrated molecular data into Patient, Observation, and DiagnosticReport resources that conform to the HL7 Genomics Reporting implementation guide, making the output immediately ingestible by compliant EHR systems without further re-encoding.

11.3 Reproducibility and FAIR Compliance

Reproducibility is a first-class design principle in MOCIP. Every pipeline execution produces a deterministic audit trail comprising: (i) a versioned configuration file recording all hyperparameters (KNN neighborhood size $K = 7$, PCA variance threshold $\alpha = 0.95$, batch correction parameters); (ii) SHA-256 checksums for all input and output files; (iii) a structured execution log in JSON-LD format capturing data provenance and transformation lineage. This design directly implements the FAIR guiding principles [13]: data are Findable via unique persistent identifiers, Accessible through standard REST APIs, Interoperable via FHIR and Parquet formats,

and Reusable through rich metadata and open licensing. The result is a pipeline that not only produces high-quality data but also provides the full methodological context required for independent replication—a standard that remains aspirational in much of the multi-omics literature [32].

11.4 Limitations and Future Trajectories

While MOCIP demonstrates near-linear scalability on large single-server cohorts, future iterations will explore distributed computing backends (e.g., Dask-on-Spark) to accommodate the projected growth of pan-cancer biobanks exceeding 10^5 samples. Furthermore, the integration of Explainable AI (XAI) modules will be prioritized, following the recent systematic mapping of XAI in omics [21]. Specifically, the adoption of biologically-informed Visible Neural Networks (VNNs) [22] and dedicated AI-driven interpretability frameworks for precision medicine [23] will provide clinicians with interpretable rationales for integrated biomarker predictions, further strengthening the bridge between computational research and bedside application.

Table 11: Projected development roadmap (2026–2027).

Milestone	Strategic Objective	Target KPI
Q2 2026	Methylation (850K) integration	>95% CpG coverage
Q3 2026	Dynamic D3.js Visualizer	Response <2 s
Q4 2026	OpenAPI 3.0 REST Services	99.9% Uptime
Q1 2027	Memory Optimization (Lazy Load)	−40% RAM footprint
Q3 2027	Apache Spark Distributed Engine	>100k samples
Q4 2027	FHIR R5 Semantic Alignment	Zero-error export

12 Conclusion

MultiOmicsPipeline (MOCIP) represents a principled and practically grounded contribution to the standardization of multi-modal data integration in precision oncology. Three concrete outcomes distinguish it from the existing landscape of bioinformatic tools.

First, by combining density-aware k-NN imputation, ComBat batch correction, and PCA-based intermediate fusion within a single reproducible DAG, MOCIP transforms a raw, heterogeneous TCGA-BRCA dataset (8.7% missing, batch $R^2 = 0.312$) into a fully complete, batch-corrected integrated matrix (composite quality score = 0.989, ARI = 0.84) without requiring any labeled information.

Second, the automated FHIR R4 Genomics Reporting bridge provides something that no comparable open-source bioinformatics framework currently offers: a clinically actionable export that can be directly ingested by EHR systems and clinical decision support platforms. This is not merely a format conversion—it involves semantic mapping through the HL7 Genomics Reporting implementation guide, ensuring that findings are interpretable by clinicians who have no exposure to bioinformatics file formats.

Third, the rigorous FAIR compliance design—SHA-256 checksums, versioned configurations, JSON-LD provenance logs—addresses the reproducibility deficit that continues to hamper translational multi-omics research [13, 32]. As multi-omics cohorts grow toward

biobank scales, the principles embedded in MOCIP will remain relevant regardless of whether the underlying algorithms evolve toward deeper architectures [25, 26] or more interpretable graph-based representations [29].

Future work will extend MOCIP along three trajectories: integration of 850K methylation arrays, deployment on a distributed Spark backend for > 100,000 samples, and the addition of an XAI explainability layer that surfaces pathway-level rationales for integrated biomarker predictions alongside each FHIR export.

Data and Code Availability

The source code for the MultiOmicsPipeline (MOCIP), including pre-processing scripts, integration modules, and FHIR export utilities, is open-source and publicly available on GitHub at: <https://github.com/ZitouniNidhal/Multi-Omics-Clinical-Integration-Pipeline-MOCIP>.

Acknowledgments

We acknowledge the TCGA Research Network and the NCI Genomic Data Commons for providing the public datasets used in this work. We also thank the Faculty of Sciences of Monastir for their institutional support. This project was conducted under the supervision of Prof. Mohsen Maraoui (Academic Year 2025–2026).

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